Speaker Notes

Student Success Analytics and Early Warning System

Slide 1: Student Success Analytics and Early Warning System

Speaker: Aritri

Good morning. We are Aritri Saha and Aston Rodrigues. Our project is titled Student Success Analytics and Early Warning System. We are developing an explainable decision support system that uses early academic and learning activity to help faculty identify students who may need support. The system assists faculty, while people remain responsible for every final decision.

Slide 2: Research Problem and Motivation

Speaker: Aritri

I will begin with the problem that motivated this project. Many institutions already collect useful student data, but signs of difficulty can remain unnoticed until an important assessment or final result. Our research explores how analytics can support earlier and more informed faculty attention.

Slide 3: The Challenge of Timely Intervention

Speaker: Aritri

The first challenge is late identification. A student may struggle for several weeks before the pattern becomes clear. The second challenge is the time required to review many separate records manually. Our system combines available early signals, such as missed assessments, low marks and reduced learning-platform activity. It then raises a support alert for faculty review. The alert does not state that the student will fail, and it should never be used as a punishment. Its purpose is to create an earlier opportunity for a helpful conversation.

Slide 4: Objectives and Data Profile

Speaker: Aritri

The next section explains our research objective and the dataset used for the prototype. We focus on information available early in a course so that any faculty response can still be useful.

Slide 5: Data Profile and Day-40 Snapshot

Speaker: Aritri

We use the Open University Learning Analytics Dataset, also called OULAD. One record in our prepared data represents one student in one module presentation. We create a snapshot at Day 40, which means the model only receives information available up to that point. The selected features include submitted, missing and late assessments, available scores, VLE clicks, active days, recent 14-day activity, studied credits and previous attempts. OULAD does not contain classroom attendance, GPA or backlog fields, so we do not claim to use them in this prototype.

Slide 6: System Workflow and Architecture

Speaker: Aritri

The workflow begins by combining assessment records and VLE activity up to Day 40. We clean and aggregate the records into one snapshot for each student and module. Machine learning models then estimate the probability of an adverse academic outcome. We calibrate the probabilities before converting them into support tiers. SHAP provides feature contributions that help faculty understand the model output. Faculty then verify the record, consider information that the model cannot see, and decide whether a supportive intervention is appropriate.

Slide 7: Model Comparison and Evaluation

Speaker: Aston

We compare Logistic Regression and Decision Tree models with ensemble methods such as Random Forest and XGBoost. We do not select a model using accuracy alone. Our main priority is recall for the High-risk group because missing a student who needs support is important. We also examine precision, PR-AUC, ROC-AUC, probability calibration and the number of alerts that faculty would need to review. The selected model produces Low, Medium and High support tiers. SHAP shows which recorded factors influenced each estimate, but it does not prove the cause of a student's difficulty.

Slide 8: Dual Dashboard Ecosystem

Speaker: Aston

The system has two dashboard views. The faculty view helps staff filter students by module and support tier, review contributing factors, and see possible support actions. Every alert requires human verification. The student view presents learning records, such as submission averages and recent activity, together with links to relevant support resources. We avoid displaying a harmful risk label or an automated judgement of ability to students. The dashboard should support communication rather than create fear or stigma.

Slide 9: Preliminary Prototype Performance

Speaker: Aston

These results come from a preliminary test using historical OULAD 2014J records. High-risk recall is 45.5 percent, which means the highest tier still misses many students with adverse outcomes. When Medium and High tiers are combined, recall increases to 74.5 percent. This is nearly three out of four cases in the combined review group. The result also increases the number of students that faculty must review, so recall and workload must be considered together. These findings describe the public historical dataset only. They do not prove that the model will perform in the same way at our institution.

Slide 10: Responsible AI and Governance

Speaker: Aston

The final part of the presentation focuses on responsible use. Educational predictions can affect how people view students, so privacy, fairness, transparency and human responsibility must be part of the system design from the beginning.

Slide 11: Responsible Use Principles

Speaker: Aston

Faculty must verify every alert before taking action. The system has no authority to change grades, enrolment, course access or academic standing. Access to student-level data should be restricted to authorised staff, and only necessary information should be collected. We will monitor performance across relevant student groups to identify unfair differences. SHAP helps explain the statistical influence of features, but it cannot diagnose personal circumstances or establish causation. Faculty should use explanations to begin a respectful discussion, not to label the student.

Slide 12: Project Roadmap and Next Steps

Speaker: Aston

Our next step is to compare predictions at Day 20, Day 40 and Day 60. This will show the trade-off between acting earlier and having more information. We will refine the risk thresholds to improve recall while keeping the review workload manageable. We also plan to extend subgroup fairness and probability calibration analysis. Finally, we will use the completed experiments and evidence to develop the research paper in the required IEEE format.

Slide 13: Questions and Discussion

Speaker: Aston, followed by both presenters

To conclude, our project aims to help faculty notice possible support needs earlier by organising existing academic and learning activity. The model provides an estimate and explanation, while faculty retain responsibility for understanding the situation and choosing a supportive response. Thank you for your attention. We are ready to answer your questions.